

Problem Set 2

Due on Canvas at noon Friday October 2nd

Instructions: **No partial credit will be awarded.** All answers must be entered in the Problem Set 2 Quiz on Canvas. Note that it is not possible to edit your responses on Canvas after submitting. **Best practice is to complete the problem set on paper and then transfer solutions to Canvas.** You may collaborate on this problem set with other students in your learning team, but each student must submit his or her own solutions.

Grading: Grading is done on a ten point scale across all problems in the problem set. **Note that after submitting on canvas, you will see a score, but that does not include any points for any qualitative questions.** Grades will be finalized a few days after the due date.

Material covered: This problem set covers lectures 6-8.

Problem 1. Tariffs and Import quotas on vegetable fiber

- a. Consider the market for vegetable fiber in the US. The domestic demand for vegetable fiber is given by

$$Q_D = 40 - 2P$$

where Q_D is quantity measured in millions of pounds and P is the price per pound. Vegetable fiber is also produced domestically in the US and the domestic supply curve is given by

$$Q_S = \frac{2}{3}P$$

where Q_S is US output in millions of pounds and P is the domestic price per pound.

Solve for the domestic (autarky) equilibrium price and quantity of vegetable fiber.

Price: _____

Quantity: _____

Solution: $40 - 2P = 2/3P \rightarrow P = 15$ and $Q = 10$.

- b. Vegetable fiber trades in a highly competitive world market at a price of \$9 per pound. Suppose the US opens up to free trade and there is enough availability that the US can import as much as it needs at this price. Do you think US would import vegetable fiber? If so, write the quantity that will be imported. If not, enter "0".

Quantity imported: _____

Solution: Demand would be 22, supply would be 6, so 16 would be imported.

- c. Subsequently, the US government imposed a \$9 import tariff. Under this tariff, what do you predict is the domestic price of vegetable fiber?

US domestic price in regime with import tariffs: _____

Solution: This would raise the effective world price to \$18, and the US would no longer import any fiber. The price would be the autarky price of 15. (They would not become an exporter, as the true world price remains at 9).

- d. What would be the total tariff revenue collected by the Government?

Tariff revenue: _____

Solution: 0, since none is imported.

- e. Without any further computations, can you predict who would be the gainers and losers from such a tariff?

Relative to free trade, US domestic producer surplus under the import would ...

- ☐ Increase
- ☐ Decrease
- ☐ Unclear without more information

US domestic consumer surplus (under the tariff vs free trade) would ...

- ☐ Increase
- ☐ Decrease
- ☐ Unclear without more information

Total US domestic surplus (under the tariff vs free trade) would ...

- ☐ Increase
- ☐ Decrease
- ☐ Unclear without more information

Why did you conclude this about total surplus? _____

Answer. US domestic producers would be better off under the tariff, US consumers would be worse off, and total surplus would unambiguously decrease as the tariff would introduce deadweight loss.

- f. (Challenge question, diagram helps here): Under pressure from trading partners, the US is considering elimination of the tariff. In its place, the proposal is to limit imports into the US to 8 million pounds a year ($Q = 8$). Under this proposal, what would you predict to be the equilibrium domestic price of vegetable fiber (understanding that the quota would lead to upward price pressure if the quota is binding)? What will be the amount produced by domestic producers?

Answer.

To figure out domestic price, note that demand must equal supply, and the supply will be met partly by domestic producers and partly by world producers (up to 8 million pounds).

So in equilibrium under this export quantity constraint, we will have domestic demand outstrip domestic supply by 8 million pounds.

Domestic price: _____

Quantity produced by domestic suppliers: _____

$$40 - 2P = 8 + \frac{2}{3}P \rightarrow P = 12$$

$$\text{Domestic } Q = 8$$

Problem 2. Hard Seltzer

The latest craze among MBA students is “hard seltzer” drinks such as White Claw and Truly that are lightly flavored and low-calorie alcoholic beverages. An enterprising SOM alum is hoping to get in on the fast-growing market by producing and selling her own hard seltzer, “Blue Paw,” in the New Haven area.

She has secured a vacant former brewery and faces monthly fixed costs of \$2,500. She estimates her variable costs, as a function of the quantity of cases of Blue Paw produced q , is given by:

$$VC(q) = 2.5q + 0.01q^2$$

- a) Suppose the prevailing price in the New Haven area for case of hard seltzer is \$15. How many cans will she choose to produce in a month? What are her monthly profits?

Quantity supplied: _____

Monthly Profits: _____

Solution: Take the derivative of VC to get MC; this is the firm’s supply curve. Set $P=15$:

$$MC(q) = 2.5 + 0.02q$$

$$15 = 2.5 + 0.02q$$

$$q = 625$$

$$\begin{aligned}\pi &= 15 * 625 - 2500 - 2.5 * 625 - 0.01(625^2) \\ &= 1,406.25\end{aligned}$$

- b) The popularity of these drinks has attracted a large number of entrants to the hard seltzer market, putting downward pressure on price. The entrepreneur has noticed that all of the entrants are offering very similar products and have similar production capabilities to Blue Paw. At what prevailing price for a case of hard seltzer would she stop producing in the short run?

Solution: look at the min(AVC), a price of \$2.50 per case.

- c) The market for the drinks stabilizes over the long run as the market matures. What do you expect the long run price to be for a case of hard seltzer to be?

Solution: look at $\min(ATC)$, where ATC equals MC :

$$2.5 + 0.02q = \frac{2500}{q} + 2.5 + 0.01q$$

$$0.01q = \frac{2500}{q}$$

$$q = 500$$

$$p = 12.5$$